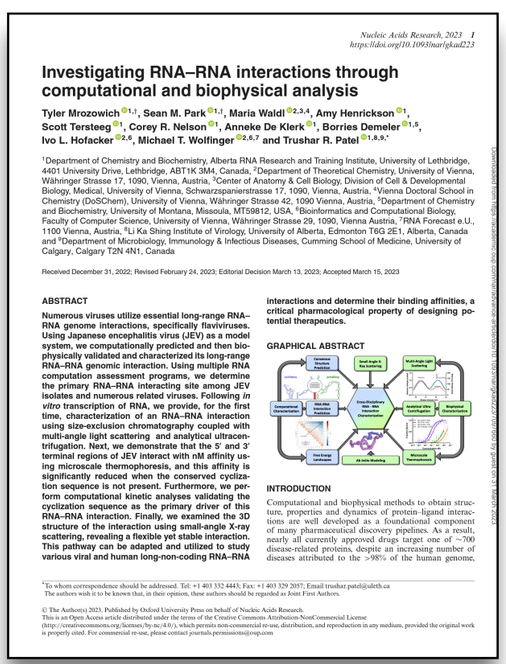
Investigating RNA-RNA Interactions Through Computational and Biophysical Analysis

Tyler Mrozowich, Sean M. Park, Maria Waldl, Amy Henrickson, Scott Tersteeg, Corey R. Nelson, Anneke De Klerk, Borries Demeler, Ivo L. Hofacker, Michael T. Wolfinger, Trushar R. Patel

Figures and title from :
Investigating RNA-RNA interactions through computational and biophysical analysis
Mrozowich, T., Park, S.M. Waldl, M., Henrickson, A., Tersteeg, S., Nelson, C.R., De Klerk, A., Demeler, B., Hofacker, I.L., Wolfinger, M.T., and Patel, T.R.
Nucleic Acids Res. gkad223 (2023)
doi: 10.1093/nar/gkad223



Nucleic Acids Research, 2023 1
https://doi.org/10.1093/nar/gkad223

Investigating RNA–RNA interactions through computational and biophysical analysis

Tyler Mrozowich [1,†], Sean M. Park [1,†], Maria Waldl [2,3,4], Amy Henrickson [1], Scott Tersteeg [1], Corey R. Nelson [1], Anneke De Klerk [1], Borries Demeler [1,5], Ivo L. Hofacker [2,6], Michael T. Wolfinger [2,6,7] and Trushar R. Patel [1,8,9,*]

[1]Department of Chemistry and Biochemistry, Alberta RNA Research and Training Institute, University of Lethbridge, 4401 University Drive, Lethbridge, ABT1K 3M4, Canada, [2]Department of Theoretical Chemistry, University of Vienna, Währinger Strasse 17, 1090, Vienna, Austria, [3]Center of Anatomy & Cell Biology, Division of Cell & Developmental Biology, Medical, University of Vienna, Schwarzspanierstrasse 17, 1090, Vienna, Austria, [4]Vienna Doctoral School in Chemistry (DoSChem), University of Vienna, Währinger Strasse 42, 1090 Vienna, Austria, [5]Department of Chemistry and Biochemistry, University of Montana, Missoula, MT59812, USA, [6]Bioinformatics and Computational Biology, Faculty of Computer Science, University of Vienna, Währinger Strasse 29, 1090, Vienna Austria, [7]RNA Forecast e.U., 1100 Vienna, Austria, [8]Li Ka Shing Institute of Virology, University of Alberta, Edmonton T6G 2E1, Alberta, Canada and [9]Department of Microbiology, Immunology & Infectious Diseases, Cumming School of Medicine, University of Calgary, Calgary T2N 4N1, Canada

Received December 31, 2022; Revised February 24, 2023; Editorial Decision March 13, 2023; Accepted March 15, 2023

ABSTRACT

Numerous viruses utilize essential long-range RNA–RNA genome interactions, specifically flaviviruses. Using Japanese encephalitis virus (JEV) as a model system, we computationally predicted and then biophysically validated and characterized its long-range RNA–RNA genomic interaction. Using multiple RNA computation assessment programs, we determine the primary RNA–RNA interacting site among JEV isolates and numerous related viruses. Following *in vitro* transcription of RNA, we provide, for the first time, characterization of an RNA–RNA interaction using size-exclusion chromatography coupled with multi-angle light scattering and analytical ultracentrifugation. Next, we demonstrate that the 5′ and 3′ terminal regions of JEV interact with nM affinity using microscale thermophoresis, and this affinity is significantly reduced when the conserved cyclization sequence is not present. Furthermore, we perform computational kinetic analyses validating the cyclization sequence as the primary driver of this RNA–RNA interaction. Finally, we examined the 3D structure of the interaction using small-angle X-ray scattering, revealing a flexible yet stable interaction. This pathway can be adapted and utilized to study various viral and human long-non-coding RNA–RNA interactions and determine their binding affinities, a critical pharmacological property of designing potential therapeutics.

GRAPHICAL ABSTRACT

INTRODUCTION

Computational and biophysical methods to obtain structure, properties and dynamics of protein–ligand interactions are well developed as a foundational component of many pharmaceutical discovery pipelines. As a result, nearly all currently approved drugs target one of ∼700 disease-related proteins, despite an increasing number of diseases attributed to the >98% of the human genome,

*To whom correspondence should be addressed. Tel: +1 403 332 4443; Fax: +1 403 329 2057; Email trushar.patel@uleth.ca
†The authors wish it to be known that, in their opinion, these authors should be regarded as Joint First Authors.

© The Author(s) 2023. Published by Oxford University Press on behalf of Nucleic Acids Research.
This is an Open Access article distributed under the terms of the Creative Commons Attribution-NonCommercial License (http://creativecommons.org/licenses/by-nc/4.0/), which permits non-commercial re-use, distribution, and reproduction in any medium, provided the original work is properly cited. For commercial re-use, please contact journals.permissions@oup.com

michaelwolfinger.com

Published under a Creative Commons Attribution 4.0 International License

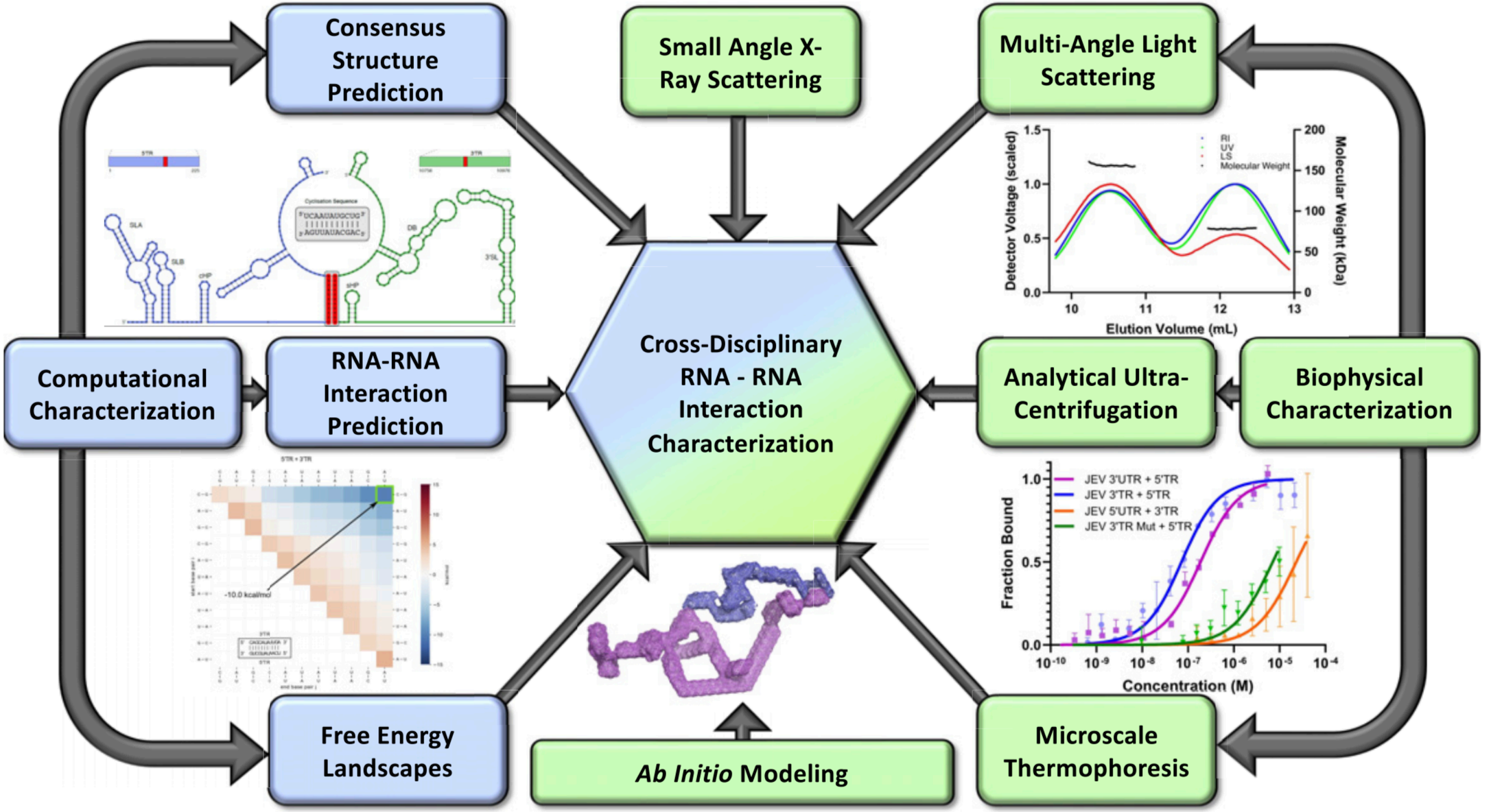


Figure from doi:10.1093/nar/gkad223

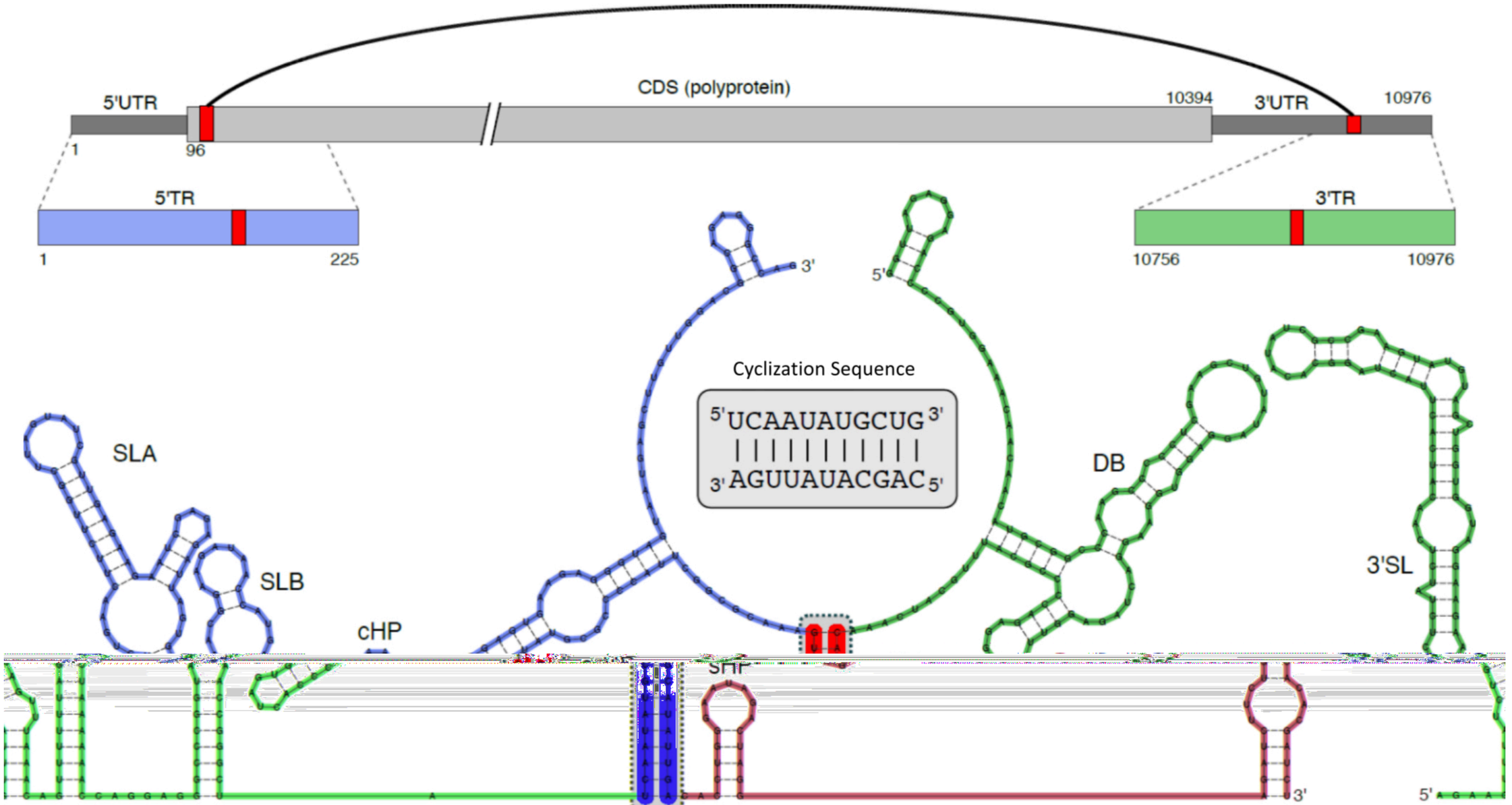


Figure from doi:10.1093/nar/gkad223

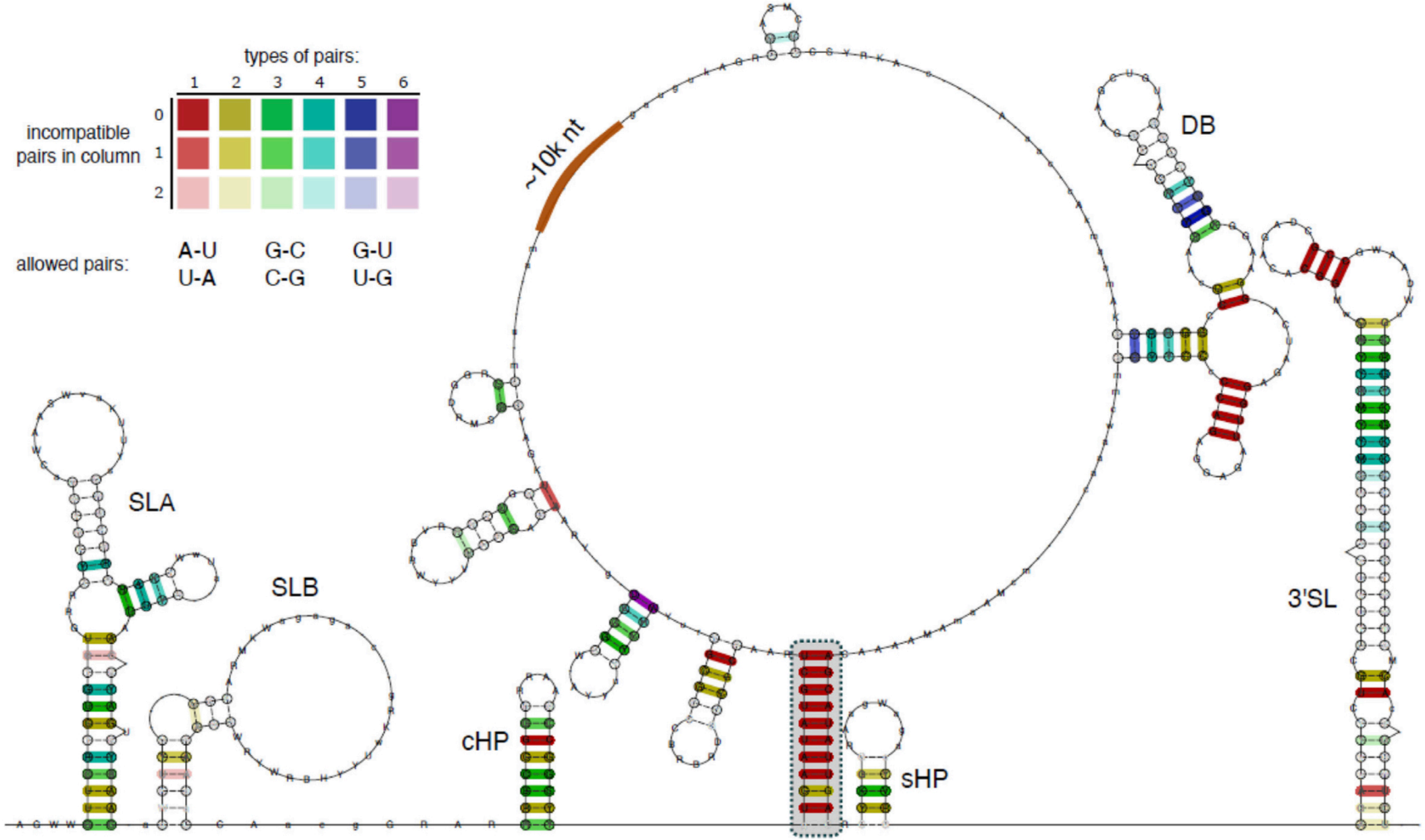
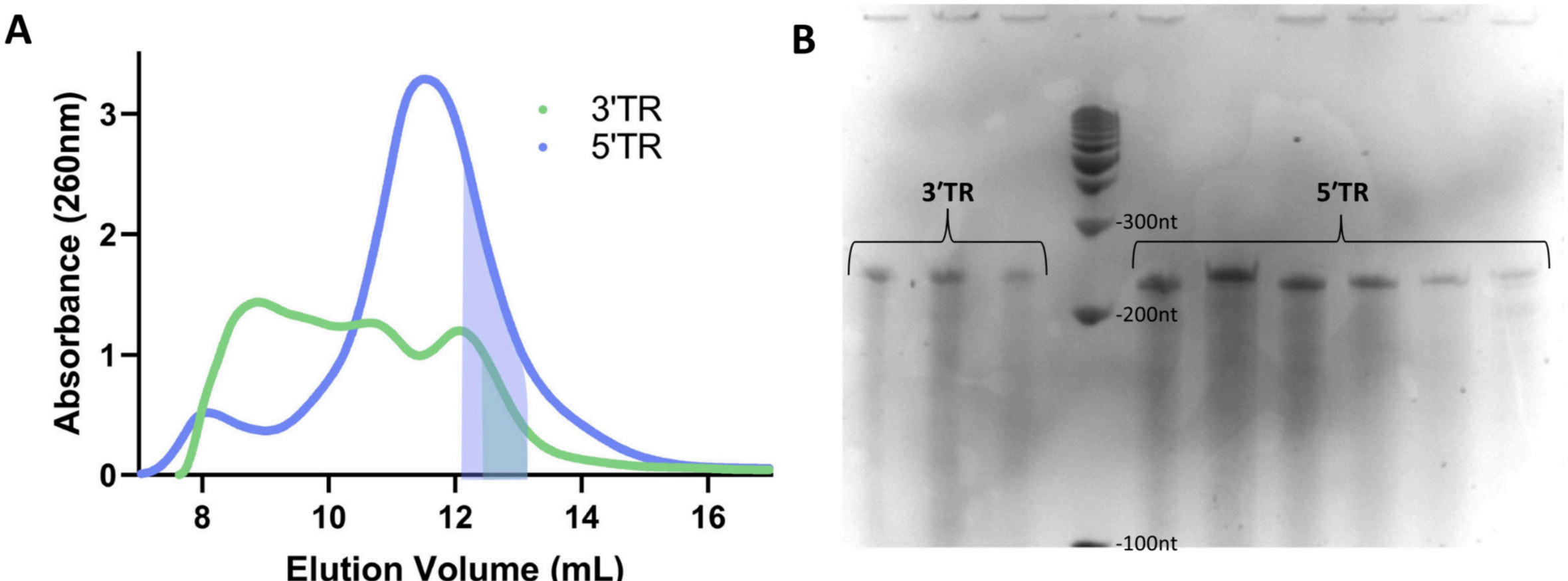
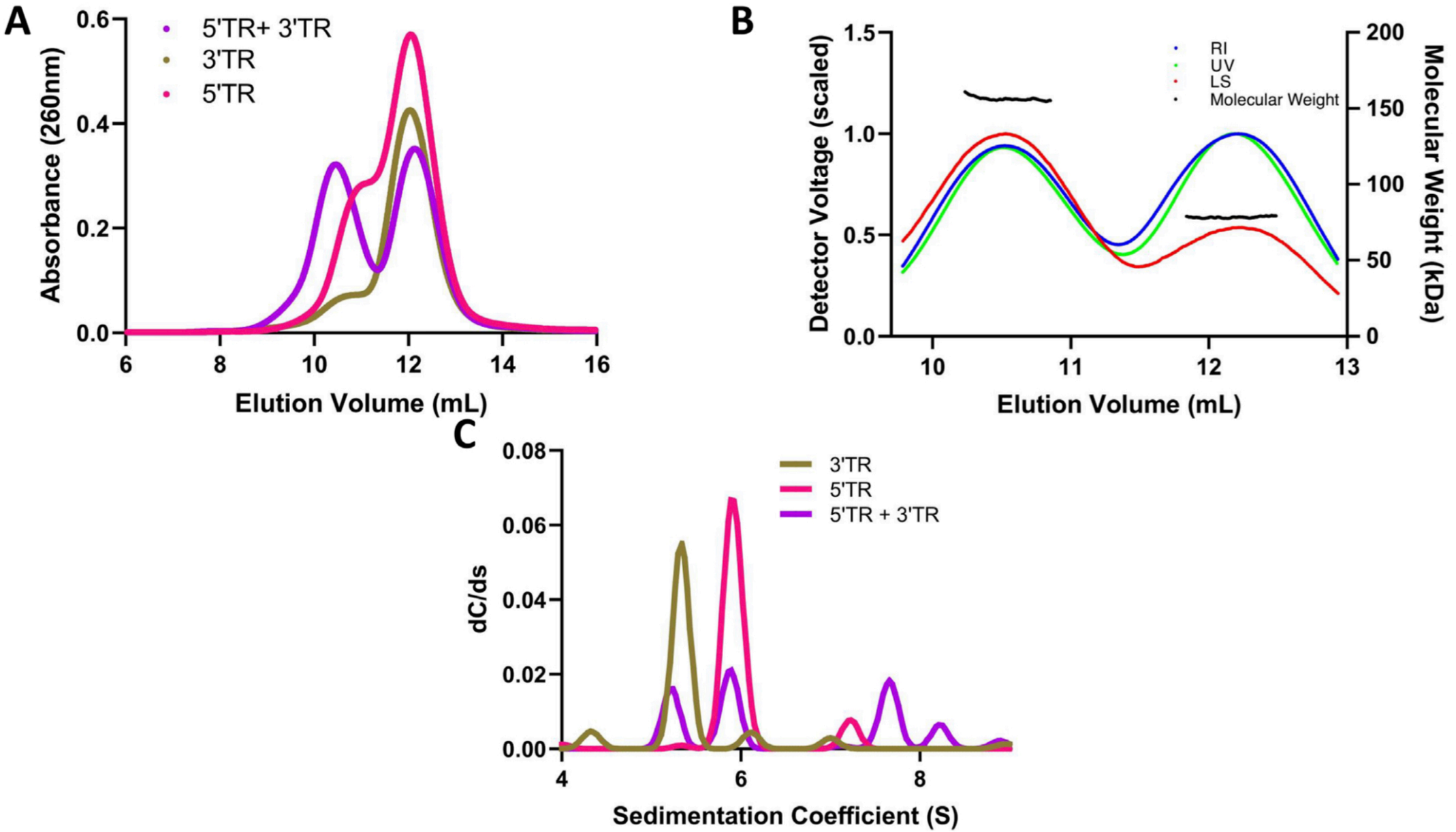


Figure from doi:10.1093/nar/gkad223



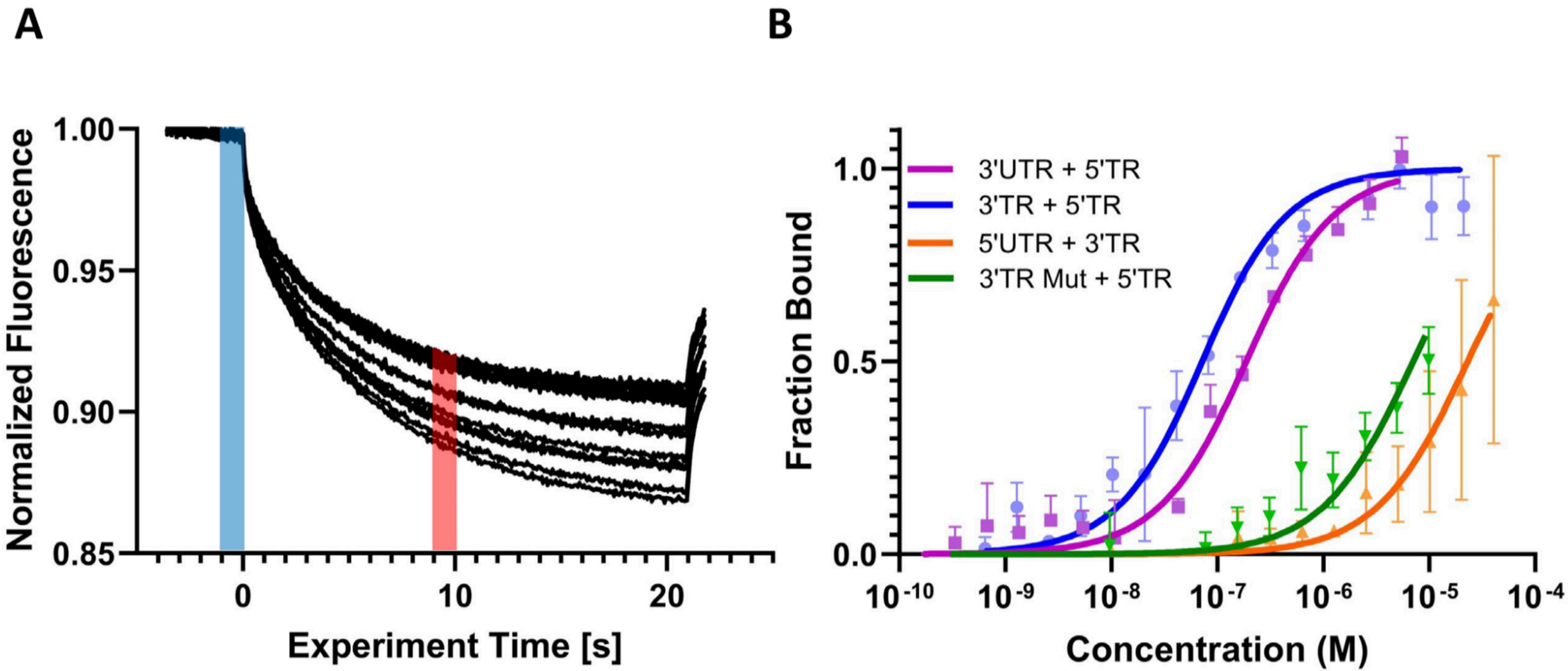
michaelwolfinger.com

Figure from doi:10.1093/nar/gkad223



michaelwolfinger.com

Figure from doi:10.1093/nar/gkad223



michaelwolfinger.com

Figure from doi:10.1093/nar/gkad223

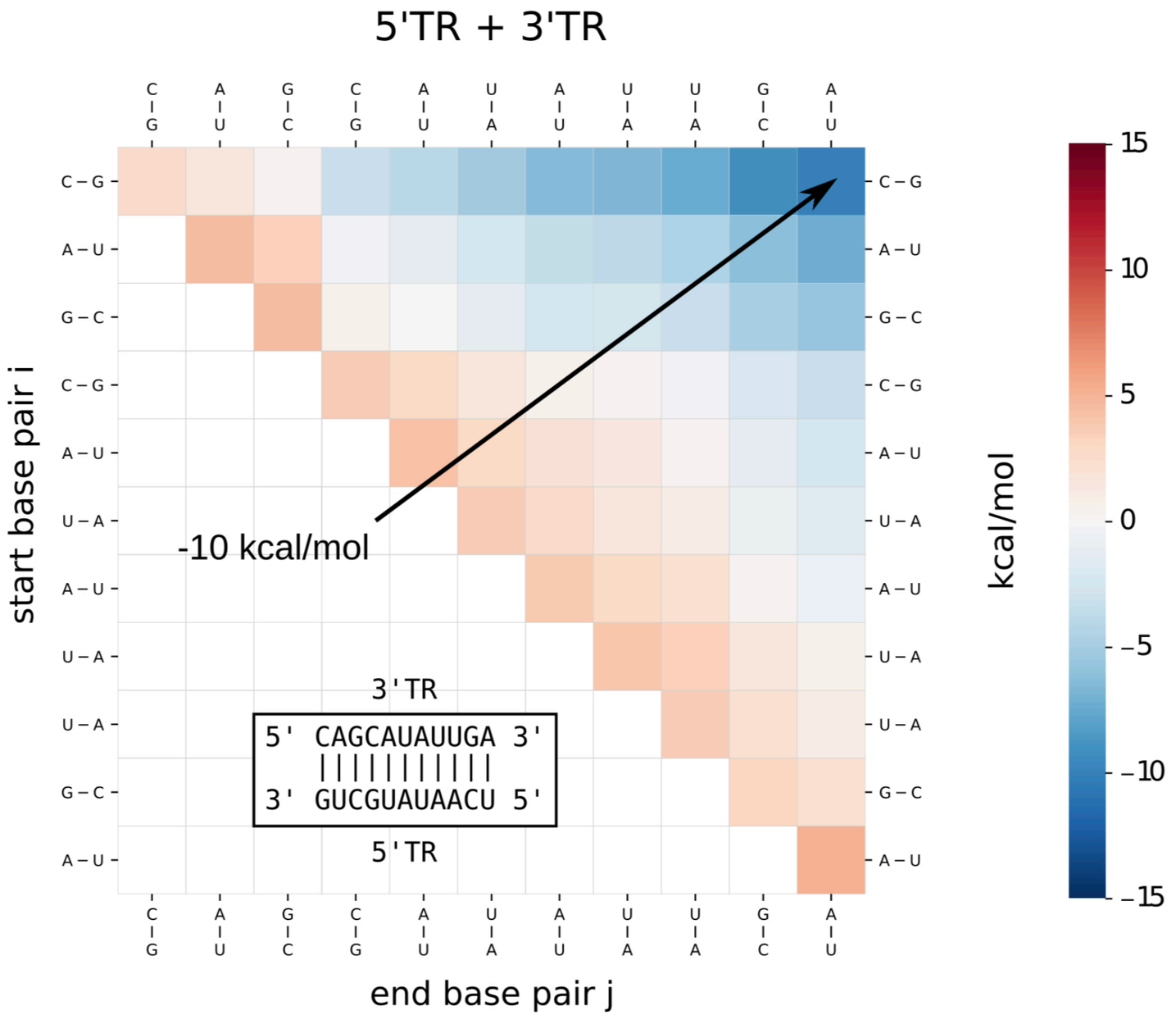
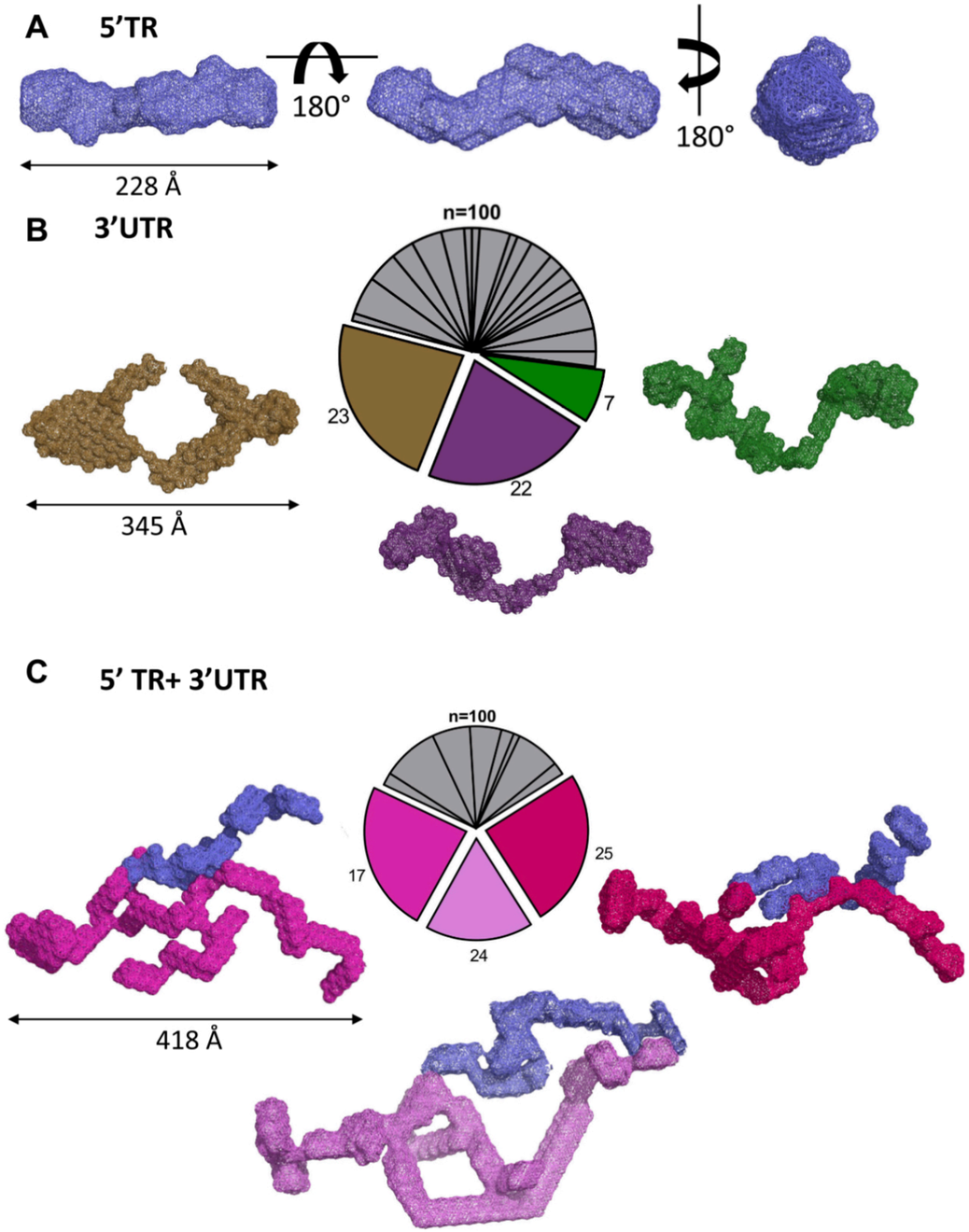


Figure from doi:10.1093/nar/gkad223



michaelwolfinger.com

Figure from doi:10.1093/nar/gkad223